

4.4.4: How Strong is your password?

Now that you've had time to play around with the password strength tester, write down your answers to the following questions. **For 2 through 5, include how long it would take to guess the password.**

1. Why do you think websites commonly require you to have at least 8 characters, and a mix of uppercase characters, lowercase characters, numbers, and symbols?
2. Try making 8 character passwords using only lowercase alphabet characters (a-z). What is the strongest password you can make?
3. Now try mixing upper and lowercase characters, still only 8 characters total. Now, what is the strongest password you can make?
4. Now try adding numbers and symbols, still only 8 characters total. Now, what is the strongest password you can make?
5. Now, using only lowercase characters, make a password that is 20 characters or longer. Is this password better or worse than an 8 character password with letters, numbers, and symbols?
6. As you play around with different types of passwords, what seems to be the single factor that guarantees a strong password?

If you would like to use the password checker to help answer questions, go to <https://www.comparitech.com/privacy-security-tools/password-strength-test>.

Solution Code

Plain text

Student answers will vary.

1. Why do you think websites commonly require you to have at least 8 characters, and a mix of uppercase characters, lowercase characters, numbers, and symbols?
The more complicated a password is, the harder it will be to guess. By mixing lower and uppercase letters, numbers, and symbols, the password becomes much stronger.
2. Try making 8 character passwords using only lowercase alphabet characters (a-z). What is the strongest password you can make?
Sample response: qmpwyxzk would take 4 months to guess
3. Now try mixing upper and lowercase characters, still only 8 characters total. Now, what is the strongest password you can make?
Sample response: QmPWYxzK would take 43 centuries to guess
4. Now try adding numbers and symbols, still only 8 characters total. Now, what is the strongest password you can make?
Sample response: Q}7~Y6zK would take 1000 years to guess

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15 5. Now, using only lowercase characters, make a password that is 20 characters or longer. Is this password better?
16 Sample response: qmpwyxzkqmpwyxzkxpzl would take 3 thousand trillion years to guess
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18 6. As you play around with different types of passwords, what seems to be the single factor that guarantees a strong password?
19 Very long passwords seems to be the best factor. Even if every character is of the same type (numbers, letters, symbols, etc.)
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Starter Code

No starter code.

